AI Based Parental Control Router-
KidSafeNet

Final Year Design Project Report

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Project Title

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In partial fulfilment of the four years **B.S. Computer Engineering** requirements, **HITEC University Taxila**, Pakistan underneath the supervision of its project adviser as well as authorized by the project review board has been recognized.

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Certificate

It is certified that Noor Ul Aen Azhar, Malik Abdul Shakoor Ahmed, and Omair Amir (22-CE-007/033/015) have carried out all the work related to this project under my guidance and supervision at the Department of Computer Engineering, **HITEC University** and the work fulfills and meets the prerequisites for the award of BS Computer Engineering degree.

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Declaration

We, hereby declare that this project neither as a whole nor as a part there of has been copied out from any source. It is further declared that we have developed this project and the accompanied report entirely on the basis of our personal efforts made under the sincere guidance of our supervisor. No portion of the work presented in this report has been submitted in the support of any other degree or qualification of this or any other University or Institute of learning. If any violation is found in this FYP report, we will be liable to be punished under the plagiarism rules of HEC.

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Dedication

This thesis is wholeheartedly dedicated to our respected supervisor,Dr. Sobia Arshad, whose constant support, expert guidance, and insightful feedback made this project a reality.

We extend our sincerest gratitude to you, Dr. Sobia Arshad, for your commitment and hard work throughout our journey. Your in-depth knowledge of network security and content filtering provided a strong foundation for our research. From the development of the initial idea to the final deployment, your input played a crucial role in shaping both the direction and the depth of our work.

Your encouragement, constructive criticism, and patience motivated us through the most challenging phases. You continuously inspired us to think critically, stay focused, and uphold high standards in our research. Your confidence in what we can accomplish together empowered us to go the extra mile and reach goals that truly matter.

This achievement in my career stands as evidence to the impact of your mentor ship. Thank you for shaping our lives both academically and personally; it has truly meant the world to us, and we are thankful we were able to learn from you.

Acknowledgement

To begin with, our most credit is offered to Dr. Sobia Arshad, our respectable supervisor, because of her guidance, her technical knowledge, and firm support were the backbone of our success in completing the Final Year Project on AI Based Parental Control Router. The project was deeply shaped by her vision and dedicated commitment to excellence. Her mentoring throughout each and every aspect of the project, starting from planning and concluding with implementation, enabled us to deal with project challenges in a directed manner.

Abstract

The prevalent use of internet and short-form digital media has exposed children to inappropriate and addictive content online at a very young age. Existing parental control system mostly uses static filtering methods and failed to monitor encrypted traffic and understand context of content. This project propose an Indigenously Developed AI based parental control router "KidSafeNet" to provide intelligent real time content filtering at network level. It is implemented in Raspberry pi 5 acting as a smart wi-fi router where all connected devices uses internet. It involves machine learning, semantic AI analysis, HTTPS traffic monitor and intelligent filtering to monitor addictive (youtube shorts, tik tok, instagram reel, etc.) and harmful online content. Our KidSafeNet monitors encrypted traffic and classify content in real time and blocks inappropriate content automatically and further redirects the user to the safe content instead of blank screen. It integrates CatBoost based traffic classification and RoBERTa based content analysis to improve filtering performance and also integrated a mobile app for remote monitoring using firebase to provide remote access to the parent. It offers a flexible, scalable and user friendly solution to modern parental control problem by integration of AI, Network Security, Embedded System and Real time remote monitoring in one platform.

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LIST OF ABBREVIATIONS

LAN	Local Area Network
WAN	Wide Area Network
IP	Internet Protocol
DNS	Domain Name System
DHCP	Dynamic Host Configuration Protocol
URL	Uniform Resource Locator
GUI	Graphical User Interface
SD Card	Secure Digital Card
CPU	Central Processing Unit
USB	Universal Serial Bus
HTTP	Hyper Text Transfer Protocol
HTTPS	Hyper Text Transfer Protocol Secure
NIC	Network Interface Card
SNI	Server Name Indication
TLS	Transport Layer Security

Chapter 1

INTRODUCTION

1.1 Introduction

The internet and digital media are becoming a bigger part of kids' lives, offering cool stuff like easy access to info and fun educational games. But there's a downside too it introduces them to risky content on apps like YouTube Shorts and TikTok. These platforms hook kids with endless feeds and tailored video suggestions, potentially harming their ability to concentrate and stay productive, not to mention taking a toll on their mental health.

Current parental control tools mostly depend on basic blocks like keywords and keeping certain sites off-limits. While that sounds okay, it really isn't good enough. They usually deal poorly with today’s complex, secure internet data flows and struggle to grasp the full picture about what's actually being viewed. On top of that, syncing up across different devices is a hassle.

For this reason, we created KidSafeNet. It’s a new kind of parental control router driven by AI and aimed at spotting tricky short-form media that could be bad for kids. Built on a Raspberry Pi 5 acting as a clever Wi-Fi hub, the setup lets it watch over everything connected to the network.

What sets KidSafeNet apart is its mix of machine learning, smart content checking, and deep dive into internet traffic. This helps catch inappropriate or overly addictive stuff that slips past older systems. Plus, it can even guide kids to safer and more helpful online spots.

To boost its skills, developers use advanced traffic sorting via CatBoost and RoBERTa for better text awareness. Parents get to join in too, through an app they can use to tweak settings and check their kid’s browsing habits anytime, anywhere.

In contrast to past solutions, KidSafeNet keeps everyone protected through the whole network rather than dealing with one device at a time. Combining AI, network safety, device tech, cloud matching, and live tracking, it delivers a practical and expandable way to help keep kids safe while exploring online.

1.2 Problem Statement:

Kids nowadays get their internet stuff from super secure platforms like YouTube, TikTok, and Instagram, which serve up quick, addictive content all day long. These days, internet chat and info sharing mostly run on encrypted stuff like HTTPS and QUIC. Because of that, old methods of keeping an eye on and limiting what kids see just don't work well anymore.

Most parental controls right now rely on set rules and filters, or they look for specific words. Some go even further by restricting certain apps or needing parents to set things up manually. The big issue is that these older methods can't tell the real meaning behind the content or understand context.

Also, a lot of parental controls struggle with watching secure data flows, spotting new short-form videos on socials, or managing access across numerous gadgets. On top of that, many existing systems can't decide properly between risky stuff and helpful info either.

Parents often have to set up each app individually on every device, making management annoying. Plus, it lets sneaky kiddos find ways around the rules by jumping to another device or network altogether.

For these reasons, we need something smarter think a special parental control router at the network level. This device would keep tabs on safe and unsafe encrypted online stuff, sort out content based on its actual context, spot inappropriate or overly addictive media immediately, and make sure controls are centralized and easy to manage for every link-up. Our plan is to build one using AI and machine learning, along with real-time monitoring and redirection tricks. All in all, it'll help provide a safer browsing experience for everyone involved.

1.3 Objectives:

The main goals of this project include:

- Create an AI powered parental control system at the network level using a Raspberry Pi 5
- Monitor real-time usage of the internet in terms of traffic using DNS-based traffic monitoring as well as monitoring HTTPS URLs
- Capture and analyze encrypted web traffic using a transparent man in the middle (MitM) proxy mechanism

- Create and label a dataset using real web traffic for the purpose of intelligent classification
- Evaluate several machine learning (ML) algorithms to identify potentially addictive and harmful online content
- Use the trained catboost ML algorithm as a classifier for network traffic data at the Raspberry Pi 5 level
- Use RoBERTa-based semantic AI analysis to identify the context of online content
- Detect and control potentially addictive short-form video content (ex. YouTube shorts, Tiktok videos, Instagram reels)
- Create a multi-layer intelligent content filtering system that uses domain filtering; URL analysis; semantic scoring; and AI classification
- Create a network-based method for blocking access to unsafe content
- Create a smart redirecting system that replaces blocked content with safe/educational content
- Create a parental control app for mobile devices based on the flutter framework
- Utilizing Google Cloud services to provide centralized management tools, remote back-up of devices, backup data and log files for use by users
- Parental controls offered as a service, so parents can manage their children's online activity from any internet-capable device.

1.4 Thesis Organization:

There are six chapters in this thesis report. All the chapters address one section of the AI-based parental control system.

Chapter 1 describes the background and objectives of the study, its significance and what motivated it. The growing children's addiction to the Internet, and increasing vulnerability to offensive online content faced by children are also addressed. Current software parental control limitations and inability, and necessity of network-based intelligent filtering unwanted content are introduced. The problem statement and goals are stated as well.

Chapter 2 contains detailed literature review on existing parental control software, router-based filtering systems and recent findings in the area of AI based content filtering and analyzing HTTPS traffic. Drawbacks of existing approaches and the resulting research gap this work is designed to address are presented.

Chapter 3 detailed information about the design, the architecture and the proposed system is presented in chapter 3. Overall architecture of the proposed system, design of DNS monitoring system, design of MITM attack for capturing HTTPS traffic, construction of dataset, machine learning model employed, semantic AI model based on RoBERTa model, design of the multi-layered filtering, integration with Firebase database, design of mobile app are presented in this chapter.

Chapter 4 details of implementation and experimental results are presented in chapter 4. Development of the modules for monitoring, integration of ML model and AI model for semantic analysis, implementation of the proposed system on raspberry pi 5, real time traffic analysis, design of system to filter specific youtube channels, mechanism of redirection and other technical aspects of the system are provided in this chapter. Details of activity tracking using firebase and features of the mobile application are also described in this chapter.

Chapter 5 environmental impacts and the aspect of sustainability in the project are analyzed in chapter 5. Contributions to ethically used internet, sustainable computing, safe digital environment, and Sustainable development goals (SDGs) are also included in this chapter.

Chapter 6 this chapter presents the conclusions, results and contributions of the project. Further extensions for the system such as multi-modal AI analysis, enhanced user monitoring and scalable cloud based system are discussed in chapter 6.

1.5 Summary

The chapter brought out the increasing concern for children being exposed to online content that is detrimental to them, due to the current internet systems. This is followed by the weaknesses of the existing parental controls, which cannot be able to analyze the traffic and comprehend the context of the content. The rationale for undertaking the project, objectives and problem statement of the project are then highlighted, followed by the introduction of the proposed router, KidSafeNet, and how it works.

Chapter 2

LITERATURE RIVEW

Nowadays, kids spend way more time on online platforms because of all the popular short videos like YouTube Shorts, Instagram Reels, and TikTok. Many medical and behavioral studies show that being exposed to this type of content can actually harm how well kids can pay attention and remember things [1].

These apps and sites hook users in with endless scrolling and special suggestions that make it really easy to just keep watching. And for kids, this could mean having trouble focusing both online and off.

To handle this issue, there are different kinds of parental controls out there apps, filtering systems through your router, and tools that use AI. Yet, most of them only use basic rules like stopping certain words or sites and ask parents to adjust settings manually.

The downside is that those methods aren't too smart. They usually miss stuff that's hidden in encryption and can't figure out whether something isn't appropriate just by reading between the lines.

That's why some researchers now want to use AI and deep learning to improve content filters and protect kids better online. People have tested various new ways to watch over what's going on with encrypted data, and smart analysis of the web. But still, most ideas don't cover all layers or don't work well enough in real-time.

In this review, we'll talk about parental control tools, smart filters using AI, and how researchers deal with keeping track of safe web activity. We'll take a closer look at their limits and explain why there's a need for a new approach like the KidSafeNet system.

2.1 Parental Control Apps for Kid Controls

2.1.1 Google Family Link

Google Family Link is an application based parental control app created by Google. It can control the use of children's Android devices by placing time limits, blocking apps, tracking their location or controlling the device remotely.

However, Family Link does not make use of artificial intelligence technology for content analysis. Family Link can tell which app a kid uses but it can't figure out if the content is actually good for them or not. Also, it doesn't check on short-form videos like YouTube

Shorts or Instagram Reels at all. Plus, it has no way to judge whether that content's okay or not. [2].

2.1.2 Net Nanny

Moving on to Net Nanny, one of the first tools for parental controls around, it just blocks sites based on keywords and categories. Net Nanny blocks websites with "adult" or "violence" in them. But here's the thing – it lacks fancy AI tech. So it misses encrypted sites and can't handle short videos or social media intelligently. Plus, it's limited to English. [3].

2.1.3 Bark

Then we've got Bark. It's an AI app that monitors kids' online chats. Yet, that's all it does nothing else on the list is covered by it. It makes use of some rudimentary AI and NLP technologies to analyze emails, texts and messages in social media to find signs of bullying, suicidal tendencies, or explicit language.

Bark doesn't actually use AI or NLP to figure out what video or graphic stuff means. Also, it only partly works for short videos, meaning harmful stuff might slip through. Plus, you got a set it up on each device individually. To top it off, it might send false alarms because it can't always grasp the full context of what it's checking [4].

2.1.4 TP-Link Deco (HomeCare)

For TP-Link Deco (HomeCare), this is a hardware system built right into WiFi routers. It's good at sifting through network traffic and keeping tabs on screen time, along with stopping access to dodgy sites. While it covers a lot by handling all connected devices, it mainly uses fixed keyword and category filters. Because of this, it struggles to figure out what a site is really about. Also, it doesn't work on secure HTTPs sites and is pretty much useless when kids switch to other networks [5].

2.1.5 Circle Home Plus

Lastly, the Circle Home Plus helps out by letting parents limit how apps are used, set screen time, and filter content based on age, all through a simple app. Setting it up is simpler than with gaming routers, but it only uses fixed-category filters.So, these controls can't tell the difference between good and bad sites, and they don't block short video clips either and they're pretty pricey. As we see, most current parental control systems follow the same basic rule approach and don't use AI. Also, they struggle to understand the content of brief videos and what they mean [6]

Table 1 Comparison of Existing Parental Control Apps

System	**Type**	**Key Features**	**Limitations**
Google Family Link [2]	Software(Mobile App)	Screen-time limits, app blocking, location tracking, Mobile lock	No AI or contextual filtering; cannot detect short-form videos, No Router
Net Nanny [3]	Software(Mobile App)	Keyword and category-based filtering	Static, rule-based; No AI; No Router, cannot detect short-form videos
Bark [4]	Software(Mobile App)	Basic AI for text and chat analysis (cyberbullying, self-harm detection)	AI limited to text only; cannot analyze video/images; no Router, No network level filtering
TP-Link Deco (Homecare) [5]	Hardware + (Mobile App)	Router-level filtering only when rules will set from app, Trend Micro web reputation updates, Rule based Filtering	No AI; static categorization; cannot block short-form content
Circle Home Plus [6]	Hardware + (Mobile App)	Time limits, app filters, network monitoring, Router-level filtering only when rules will set from app	No AI; rule-based filtering; no short-video detection; costly subscription

As evident from the above comparison, most of the currently available parental control systems use rules and do not employ any AI-based technologies. In addition, all the solutions have difficulties in analysing short-form videos and comprehending their meaning. Thus, a solution involving smart, dynamic, and real-time analysis via an intelligent system is necessary.

2.2 Research-Based Developments in Parental Control and Contextual Filtering

Several recent innovations in parental control and contextual filtering use AI and context-based technology to improve static filtering tools. Canino et al.[7] suggested a prototype design for router-based parental control architecture that employs deep packet inspection in order to monitor social media activities in the network environment. While this solution allows increasing centralized management, it still does not utilize artificial intelligence for decision making and is not implemented in the real life scenario. Moreover, Sangeetha and Mythili[8] introduced a novel system using AI for the content control with age determination in order to protect children from media exposure. Using deep learning, the researchers managed to achieve 98.83% accuracy of age detection by implementing convolutional neural network. Nevertheless, the system remains untested on real-world online platforms and is still in the experimental phase.

Another research suggests employing 5G networks for privacy-preserving parental control[9]. The study suggests introducing a "smart response" instead of simple blocking together with using Private Set Intersection (PSI) and Private Set Operations (PSO) protocols for privacy preservation. The algorithm applies edge computing combined with artificial intelligence to detect potentially harmful content and apply automatic network-level parental controls. In particular, the study focuses exclusively on analyzing text and website content, while audio, video, and images remain a subject for future studies. In addition, this innovation requires 5G infrastructure support and network providers' cooperation for the implementation. Overall, the algorithm remains tested only theoretically via simulation, and there is no real-world implementation of the idea yet.

Authors [10] performed a comparative study of seven different parental control systems (three routers, two DNS resolvers, one software tool). For the experiment, each solution was tested by visiting one million domains (Tranco dataset) and measuring the number of blocked websites. Domains were classified by types (Adult, Gambling, Hate, Terrorism, etc.) according to Cisco Umbrella data. By measuring blocking consistency, over blocking, and under blocking, the researchers revealed that some solutions failed to block up to 96% sensitive websites, while others sometimes over blocked valid websites. The majority of parental control solutions used DNS-based filtering and exhibited inconsistency in blocking between different devices. Some solutions were found to be black boxes lacking transparency and context-based decision making, thus proving themselves unreliable.

Authors [11] proposed an innovative solution called TikGuard which applies deep learning in form of transformers for classifying TikTok videos into four classes: Safe, Adult, Harmful, and Suicide. TimesFormer, ViViT, and VideoMAE models were fine-tuned for video recognition using a custom dataset TikHarm. The main steps in the process involve extracting video frames, normalizing the dataset, augmenting the data, and finally performing fine-tuning of models with selected optimal parameters. Accuracy, precision, recall, and F1-scores of each model were measured to select the one with maximum accuracy (TimesFormer - ≈86.7%). Limited sample of videos (3,948 in total) hampers model generalization. Videos were considered solely based on visual cues without using any audio/text cues that could provide additional context.

There are numerous research works concerning analysis of mobile encrypted traffic with deep learning. Aceto et al. introduced an innovative approach based on deep learning for classifying mobile encrypted traffic by analyzing flow-level characteristics without having access to payload information. The study successfully proves that encrypted traffic can be effectively classified by utilizing statistical characteristics; however, the classification is limited to traffic types identification and cannot be used for content-based filtering[12]. Lotfollahi et al. presented an effective deep learning-based classifier named "Deep Packet" for detecting types of encrypted network traffic. The system automatically extracts the necessary features by analyzing raw traffic data and achieves maximum accuracy of classification with the help of neural networks. As far as it can be concluded from the study, while the classification performance is very high, Deep Packet does not perform any interpretation of the semantics of the content and cannot make real-time filtering decisions required in a parental control system[13].

Rezaei and Liu conducted a thorough study of deep learning-based methods for encrypted traffic classification[14]. The researchers point out that most of the modern approaches base their algorithms on analyzing flow-level features of packets like packet length, timing, and traffic patterns to identify application types. However, such approaches have significant limitations since they do not provide any content analysis or user behavior tracking. Velan et al. presented a comprehensive survey on encrypted traffic classification techniques, discussing statistical, machine learning, and flow-based approaches. The study emphasizes that most methods rely on traffic patterns rather than content understanding, which limits their ability to perform context-aware filtering or identify specific types of harmful content [15].Finsterbusch et al. presented a comprehensive review of existing payload-based encrypted traffic classification algorithms and methods[16]. The researchers pointed out that those approaches can provide efficient analysis of application-level traffic types; however, they do not work with encrypted traffic since the payload data is absent.

Thus, although these innovations represent great progress in developing intelligent parental control and content filtering system, most of them are still at the research/prototype stage. Additionally, none of these papers addresses the problem of combining hardware-level filtering with AI.

.

Table 2 Summary of Research-Based Developments in Parental Control and AI Filtering

S. No.	Year of Publication	Title	Methodology	Limitations
1	June 2025	A Router-Based Parental Control Tool for Safe Social Network Usage	Router-based parental control using deep packet inspection (DPI) for analyzing and managing social network traffic	Prototype only; lacks AI-driven decision-making;

2	2024	Age-Based Content Controlling System Using AI for Children	Deep Learning (CNN) for detecting user age and controlling inappropriate media content	Achieved 98.83% accuracy; still experimental; not tested on live online data
3	2021	Parental Control with Edge Computing and 5G Networks	Proposed a privacy-preserving parental control protocol using edge computing and AI in 5G networks. The system introduces “Smart Response” instead of simple blocking, and uses Private Set Intersection (PSI) and Private Set Operations (PSO) protocols for privacy. The method relies on AI classifiers deployed at the edge to detect harmful content and provide automatic, network-level parental control	Focused only on text and website content; audio, video, and image analysis are left for future work. Relies on 5G infrastructure support and assumes cooperation from network operators. Implementation is limited to simulation and cryptographic performance, without full real-world deployment.

			while preserving both child and parent privacy.	
4	2025	To Block Or Not To Block? Evaluating Parental Controls Across Routers, DNS Services, and Software	Conducted a comparative analysis of seven parental control systems (three routers, two DNS resolvers, one software tool). Tested each by visiting one million domains (Tranco dataset) and recording blocked sites. Domains were categorized (e.g., Adult, Gambling, Hate, Terrorism) using Cisco Umbrella data. Evaluated blocking consistency, overblocking, and underblocking through network	Found that systems frequently fail to block up to 96% of sensitive content and sometimes overblock legitimate websites. Filtering was mostly DNS-based and inconsistent across devices. Some solutions acted as black boxes, lacking transparency and context-aware AI, making their operation unreliable.

			measurements and packet captures.	
5	2024	TikGuard: A Deep LearningTransformer-Based Solution for Detecting Unsuitable TikTok Content for Kids	The study proposes TikGuard, a transformer-based deep learning approach for classifying TikTok videos into categories (Safe, Adult, Harmful, Suicide). It fine-tunes TimesFormer, ViViT, and VideoMAE models using a curated dataset named TikHarm. Key steps include video frame extraction, normalization, data augmentation, and model fine-tuning with optimized hyperparameters. Performance was evaluated using	Limited dataset size (3,948 videos) restricts model generalization. The dataset focuses only on visual features, excluding audio and textual content that could provide better context. Lacks multimodal fusion and cross-platform testing. Comparison with existing systems is limited due to dataset uniqueness.

			accuracy, precision, recall, and F1-score with TimesFormer achieving the best accuracy (≈86.7%).	

2.3 Limitations of Existing Solutions

From what we've read, there are some common issues that pop up. Most parental control systems use fixed rules for filtering rather than adapting to the context. Also, these systems usually can't handle encrypted HTTPS traffic or short-form videos.

A lot of the research only works in experimental settings. So while useful for studies, they don't really translate to actual apps. One team, Draper-Gil and company, looked into using traffic behavior like packet intervals and flow durations to identify different data types. While this method does let you tell one kind of traffic from another, it can't filter content based on what it actually is. This makes their approach not too helpful for more intelligent filtering tasks. [17].

2.4 Research Gap and Motivation

From the above review, we can see there's no system yet out there that does both network-level monitoring and smart machine learning analysis.

The aim here is to create an integrated parental control system. This would handle monitoring, sorting, and guiding network flows to suitable content, all based on traffic traits.

First off, Moore and Zuev put out a way to classify internet traffic types using Bayesian stats methods. While their accuracy was good, the setup doesn’t cut it for real-time choices or smart filtering [18].

Next up, Shafiq et al. looked into how users interact with mobile video apps, zeroing in on common features of today’s video services. Still, their work mostly covered the user end, and didn't fit this research's bigger picture [19].

Wan Ismail et al. [20] created a parental control system using a Raspberry Pi that does website filtering, user monitoring, and manages internet usage time at the network level. It blocks inappropriate sites with predefined rules and offers central control for connected devices. The setup is manual and doesn't use AI for content analysis. Plus, it can't classify encrypted HTTPS traffic or spot new harmful content dynamically. This limits its usefulness in today's online world.

Sahu [21] came up with an intelligent parental control system using an agent-based approach to automate restrictions and monitoring. While this boosts automation in parental controls, the method still mainly uses fixed rules and conventional means. They didn't use machine learning for sorting content on the fly. Plus, their setup doesn't analyze encrypted HTTPS data or understand context in online stuff.

KidsWatch is parental control software that shields kids from unwanted online content by blocking sites and tracking activity. Parents can use it to limit site access, watch instant messaging, and manage web time too. Plus, it sends report emails about what the kids do online. The downside? It depends on fixed filters and doesn’t handle smart threat analysis or AI for content checks [22].

CyberPatrol [23] is a parental control app for website filtering, internet scheduling, and program control, plus it lets you watch instant messaging and web actions. You can use it at home, school, libraries, and work to stop bad content and keep tabs on what folks are doing. While it has good monitoring tools, it doesn't use AI for smart content analysis or find new risky sites automatically.

2.5 Summary

In this chapter, a number of available parental control tools, routers-based filtering technologies, and latest findings about AI-enabled content filtering and encrypted traffic analysis have been considered. Several different technologies and their capabilities and weaknesses have been discussed. From the literature review, it became clear that the vast majority of existing technologies are based on static filtering mechanisms without any intelligence in understanding the nature of content. This finding served as a starting point for the identification of the research problem.

Chapter 3

METHODOLOGY

The system uses a hybrid approach which includes hardware level filtering with AI based content analysis and an interface in the form of a mobile app. The system is structured such that smart filtering is achieved at the router level, and also the parent is provided with complete control and visibility over the system through a cross platform Flutter based application.

3.1 Architecture of Proposed Design

The proposed KidSafeNet system adopts a hybrid architecture. Network level traffic inspection, content analysis with artificial intelligence, cloud synchronization and mobile based parental control features are merged into a single system. Overall, KidSafeNet provides intelligent and real time centralized parental control for all devices in network without install software in each users device.

There are 5 main components:

1. Raspberry Pi Router Module

Raspberry Pi 5 acts as main gateway to all the connected devices and the Wi-Fi router. All the devices connected to the Raspberry Pi 5 send their Internet data through it, and so all the incoming traffic can be easily monitored or filtered at the network level. Router module performs packet forwarding, traffic interception, hotspot functionality and inter-process communication.

2. Traffic Monitoring and Interception Layer

This layer is to monitor the internet traffic in real time via DNS analysis and HTTPS interception. The DNS traffic can analyze the domain names relating to encrypted connections, a transparent MITM proxy is set to intercept the HTTPS requests and captures the entire URLs and request metadata. In the meantime, this system allows us to see the traffic inside the encrypted connections, and it serves as the prerequisite for smart traffic analysis.

3. AI-Based Filtering Engine

The AI Filtering Engine is the brains of the system and makes use of machine learning classification, semantic AI analysis, and smart filtering technologies in order to find inappropriate or addicting content in real-time. It merges CatBoost-based traffic classification, keyword and URL analysis, smart signal weighting and scoring and RoBERTa-based semantic analysis of online content.

The system can detect short-form and addictive contents like YouTube Shorts, Instagram Reels, TikToks, unsafe searches and bad multimedia contents. AI can predict what the unsafe content will be and block it and policy will be automatically applied for filtering.

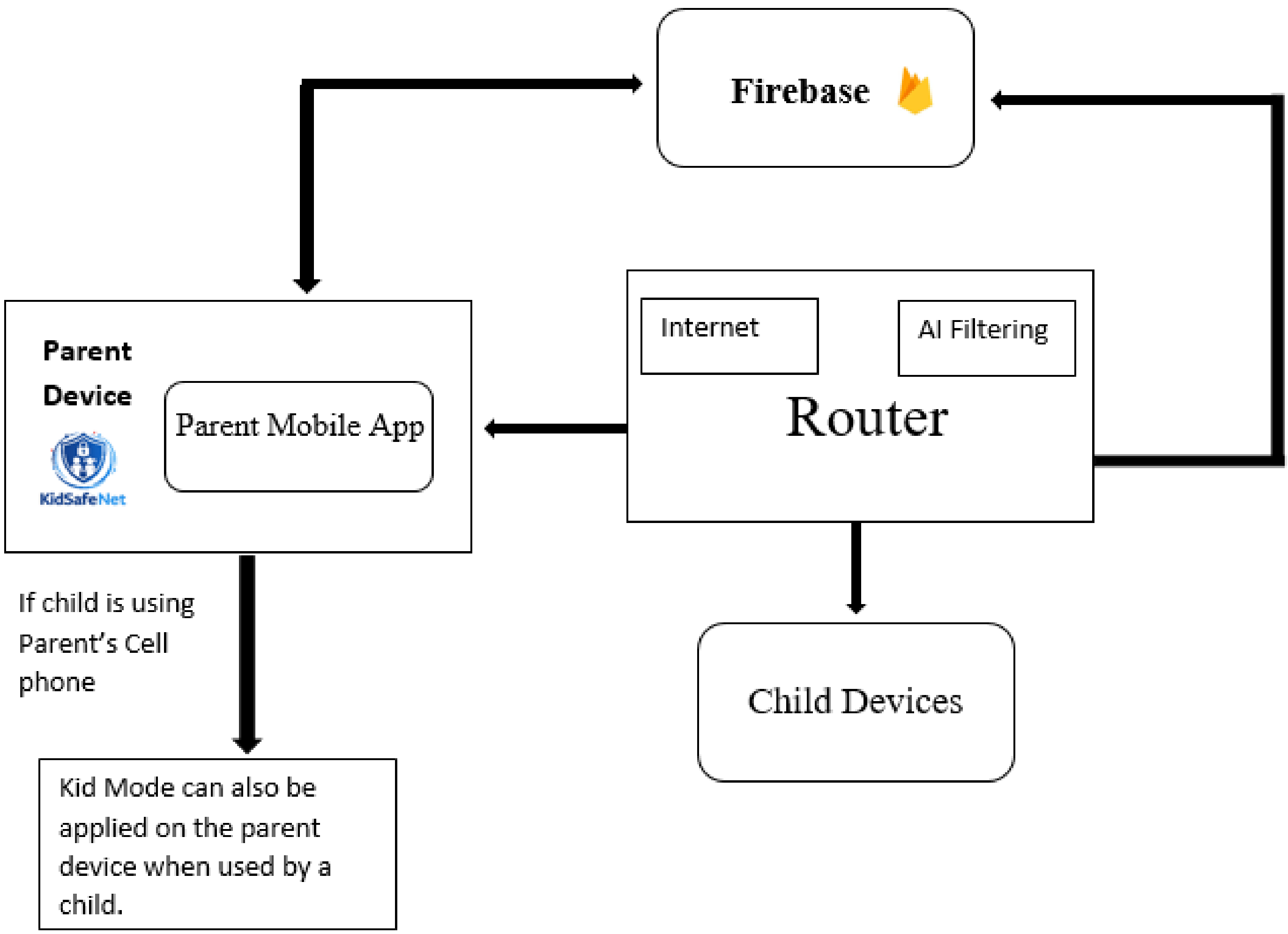
4. Smart Redirection and Safe Content System

The technology does not redirect to standard blocked or error pages but uses a smart redirection function which replaces inappropriate material with child-friendly alternatives. A tailored environment for platforms such as YouTube has been developed which prevents the display of inappropriate recommended videos, thumbnails and short-feeds by removing them and embedding useful, educational child-friendly material.

5. Mobile Application and Cloud Integration

With Firebase integrated services, we build a Flutter mobile application to remotely supervise and control the system at real-time by the parent users. The application features include control Kid mode, activity monitor, filter setup, usage statistic report. Firebase Firestore provides cloud synchronize service, logs user activities, remote config update service and real-time communication between raspberry pi router and mobile application.

This architecture will allow to have a central control using AI as parental control but will remain scalable, flexible and deployable. With the association of network security, artificial intelligence, embedded systems, monitoring by the cloud and an artificial intelligence the system will offer a smart and feasible solution for a secure internet.



Our Overall System Architecture Diagram:

Figure 1: Overall System Architecture

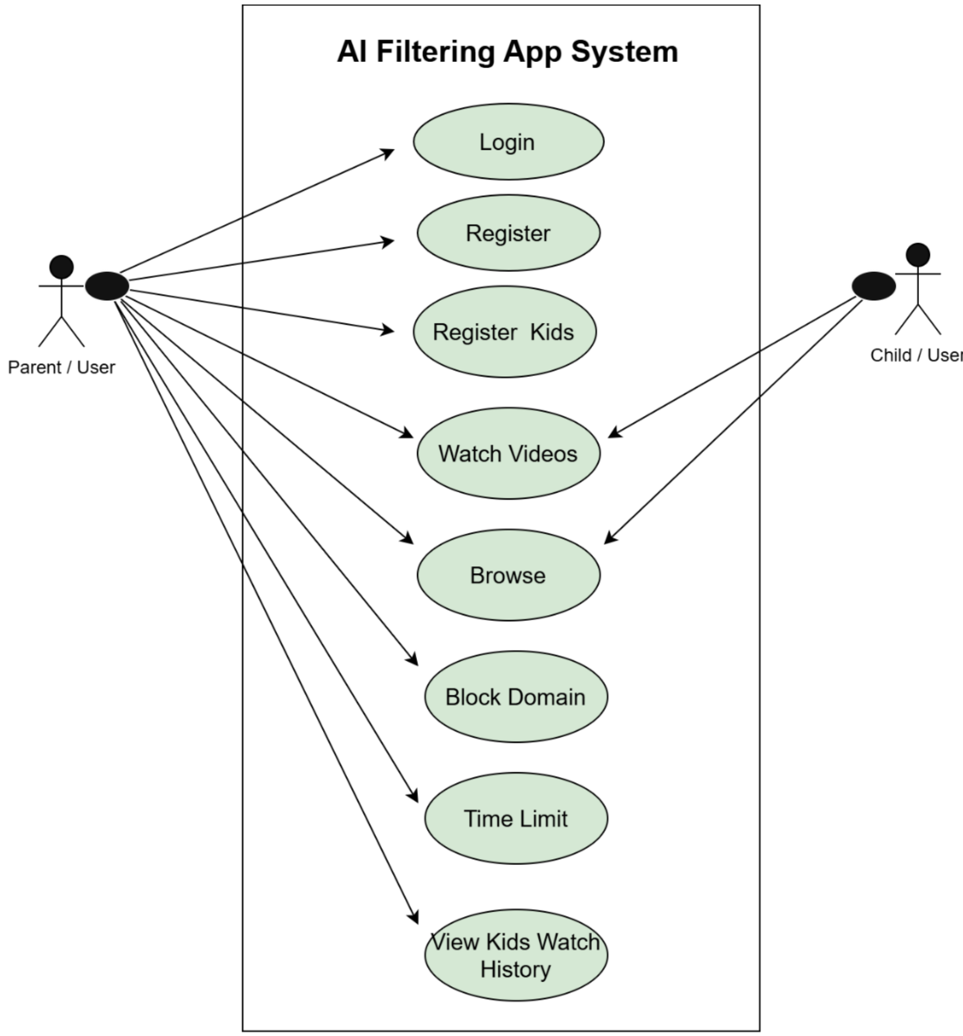


Figure 2:Use Case Diagram of Proposed System

3.2 Hardware Components:

3.2.1. Raspberry Pi 5:

Figure 3: Raspberry Pi 5

The Raspberry Pi 5 is a tiny, powerful single board computer that can handle many general purpose computing and networking jobs.In its case, a 64-bit quad-core Cortex-A76 processor, fast LPDDR4X RAM, Gigabit Ethernet, an embedded Wi-Fi and Bluetooth, various USB and HDMI ports make it well suited for embedded systems as well as more advanced networked devices. For this project, it acts as the primary central processing unit in the PiRouter, and the main component of the system; network traffic is managed here, monitoring tools such as mitmproxy are run here, a real-time dashboard is hosted here and interaction between connected devices occurs here. Its processing and networking power is ideal for implementing the real-time URL monitoring and controlling functions needed for this project.

3.2.2 Raspberry Pi 5 Case with Cooling Fan:

Figure 4: Raspberry Pi 5 Case with Cooling Fan

The Raspberry Pi 5 case is a custom made enclosure, it can provide protection for the Raspberry Pi and also the proper conditions to keep it working. This is a clip together case with built-in cooling fan that controls the speed based on the temperature of the board and ventilation slots. External hardware devices will have their corresponding mounting points within the case.It's purpose in this project is to ensure the proper operating temperature, stability, and performance of the system, which in a PiRouter, must be running a constant monitor over network, process, or have multiple devices attached to it, so that in high workload the system is working well without over heating thanks to the active fan.

3.2.3 Raspberry Pi 27W USB-C Power Supply:

Figure 5: Raspberry Pi 27W USB-C Power Supply

This Raspberry Pi 27W USB-C power supply will be the only power supply used to power the Raspberry Pi 5.The power supply has been optimized to deliver adequate voltage and current via the USB-C interface, with various output levels available which are capable of providing stable power to situations that place a very heavy demand on processing power, such as contemporary Raspberry Pi models, when applications demanding a large processing power are running. This 27W USB-C power supply is very important in this project because the PiRouter will be constantly performing network communication while having real time monitoring in the background. A good power supply will ensure that the system is always stable.

3.2.4 SanDisk Ultra 64GB microSD Card:

Figure 6: SanDisk Ultra 64GB microSD Card

High Speed Micro SD storage device for storing the operating system, programs and files needed for the Raspberry Pi. The speed characteristics of a Micro SD storage device is a very important parameter that is crucial for fast booting of the system and programs are running fast. System storage capacity: To store the Raspberry Pi OS plus some optional tools and library that required for the project. In this project the storage media will be Micro SD card. It will also be where all the configurations of the system are stored along with monitoring scripts and the dashboard. Its speed will also be important for system to be quick to respond.

3.2.5 Ethernet Cable (LAN Cable):

Figure 7: Ethernet Cable (LAN Connection)

Ethernet cable: this cable enables the connection of the raspberry Pi to the external network or an internet source (a router). This cable is necessary for reliable data transfer to and from the external network.The connection is much more reliable with very less delay, unlike using wireless. Ethernet cables allow continuous bandwidth, and are more robust to external interference, so they fit perfectly for networking systems which need to send data continually. In this project, it is used to connect to the main internet source and make the system into a PiRouter.

3.3 Software Components:

3.3.1 RealVNC Viewer:

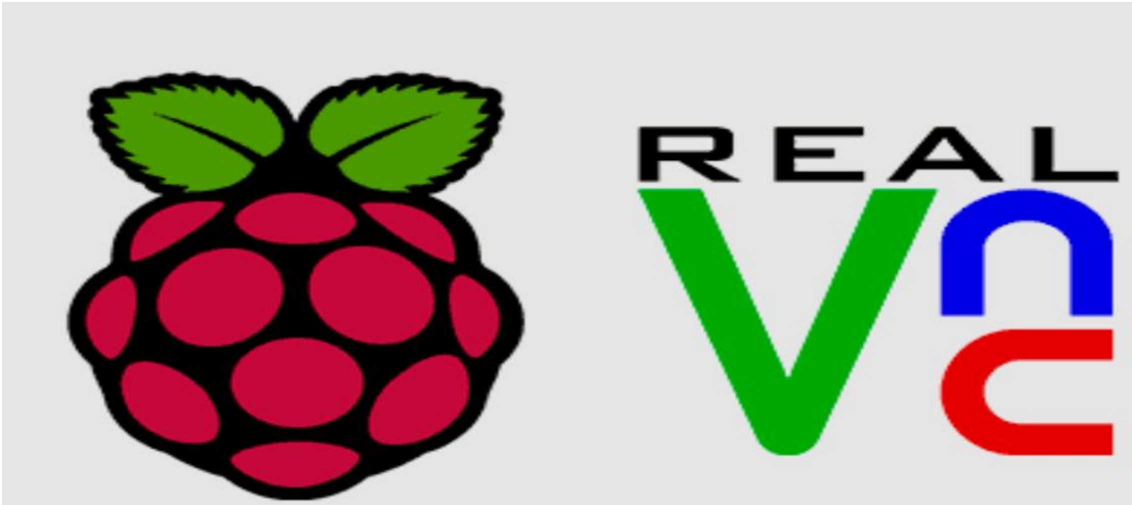


Figure 8: RealVNC Viewer Interface

RealVNC Viewer is one kinds of remote desktop software which allows one user to remotely control another computer via the network.It presents one graphical interface, and enables a user to interact with the Raspberry Pi in the exact same way it would feel to directly connected to the Pi with its peripherals. The application is based on the Virtual Network Computing technology and allows remotely monitor, configure, and manage the system without any monitor, keyboard or mouse, so in this case we use it to log into the Raspberry Pi which is currently running our PiRouter setup and configure the network, monitor its processes and view the URL monitoring dashboard easily from our local desktop, and in an invisible way due to absence of the monitor.

3.3.2 Python

Python is used as the primary programming language for network monitoring, packet analysis, AI integration, and backend system development.

3.3.3 mitmproxy

mitmproxy is used to implement transparent HTTPS interception and URL monitoring. It enables extraction of complete HTTPS URLs and request metadata after TLS decryption.

3.3.4 Flutter

Flutter is used to develop the KidSafeNet mobile application. It provides a cross-platform user interface for Android and other supported devices.

3.3.5 Firebase

Firebase services are used for authentication, Firestore database synchronization, activity logging, and real-time remote monitoring.

3.3.6 PyTorch

PyTorch is used for implementing and running the RoBERTa semantic AI model for contextual content analysis.

3.3.7 CatBoost

CatBoost is used as the primary machine learning classifier for real-time traffic classification.

3.3.8 Google Colab

Google Colab is used for dataset preprocessing, machine learning model training, testing, and evaluation.

3.4 Proposed System Methodology

KidSafeNet, which we propose, implements the multilayer intelligent filtering technique including the network traffic monitor, the machine learning technique, semantic AI technique, router-level implementation, and real time control by parents into a whole framework. The methodology starts with monitoring and intercepting the network traffic and then extract feature and perform AI based analysis, can finally allow or block or redirect the web pages in real-time according to the decision of filtering, logs and control also be synchronized via the firebase cloud.

General working of system includes DNS monitoring, HTTPS url interception, building dataset, ML classification, semantic AI classification, router level filtering, smart redirect, firebase syncing and Mobile app control.

3.4.1 Traffic Monitoring Layer

The traffic monitoring layer has the responsibility to monitor and log traffic that goes through the Raspberry Pi router. This provides an opportunity to analyze encrypted/non-encrypted traffic and builds a platform on top of it for intelligent AI based filtering. The monitoring system has two main phases which are the:

- DNS Based Network Traffic Analysis
- URL Monitoring via MITM Proxy

These monitoring module produce data necessary for machine learning and real-time filtering.

3.4.1.1 DNS-Based Network Traffic Analysis

In the first monitoring phase, we deploy the DNS based traffic analysis system using python with Scapy library. It captures packets directly from the interface selected and runs in promiscuous mode. Analyze and extract domain names along with the IP addresses by inspecting DNS packet.

We implement a DNS cache in order to save the mapping between domains and IPs. This cache is later used to determine the domains behind the encrypted HTTPS and QUIC traffic. We try to extract SNI data from TLS handshakes whenever it's available. All the observed packet information like IP addresses, protocols and identified domains is displayed in the real-time through the GUI developed using Tkinter.

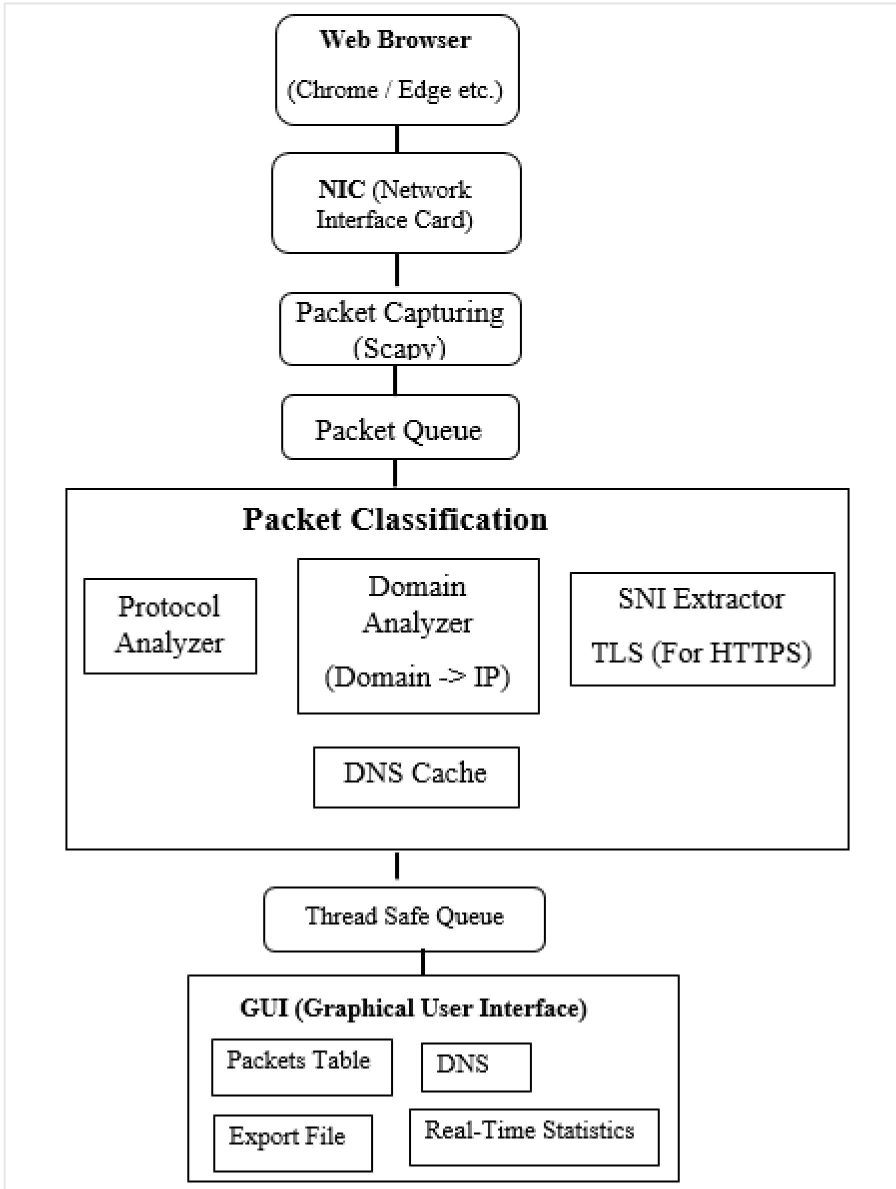


Figure 9: DNS-Based Network Traffic Analysis

Algorithm:

DNS-Based Network Traffic Analysis

- Start up the system and get all the needed libraries running for catching network packets, dealing with data, and showing it graphically. Then, pick your computer's active network card and switch on promiscuous mode. This way, you catch both incoming and outgoing network packets.

- Set up an empty DNS cache. It will hold IP addresses with their matching domain names. Keep in mind, this helps later on with quicker lookups.
- Next, start a listening process in the background. It focuses on sniffing out all the network traffic moving through the NIC. Don't worry; nothing gets changed or blocked.
- When a packet comes in, add it to a processing queue. Doing this makes sure each piece of traffic gets handled in order. Once queued, read through them individually.
- To begin analyzing a packet, first check if it contains an IP layer. If it doesn't, simply toss it. For those that do have IP layers, pull out the source and destination IPs, along with the transport protocol like TCP, UDP, or ICMP.
- Keep sifting through that packet header info. Try to spot application-level protocols, like DNS, HTTP, HTTPS, QUIC, or others. DNS packets need special care. Extract domain names from queries and responses, find the IP addresses, and pop them into the DNS cache.
- Now, onto HTTPS packets. Search for that TLS handshake message within the encrypted load. Grab the Server Name Indication from the TLS ClientHello message when you can. But, if that fails, hit up the DNS cache with the destination IP to see if a domain name pops up there.
- With protocols like HTTP or QUIC, hunt for domain names using headers and cache lookups. Packets without any trace of a domain get marked as "unknown."
- Take the collected data, put it together neatly, and feed it to the graphical user interface. This lets users keep an eye on what's flowing through in real time. Also, regularly clean out old cache entries to prevent memory overload and keep the UI updating smoothly.
- Got interested in long-term analysis? Save that logged info to file for later digging. Let the monitor run as long as the user needs it to, but once they're done, wrap things up nicely. Release all resources and power down the app without any hitches.

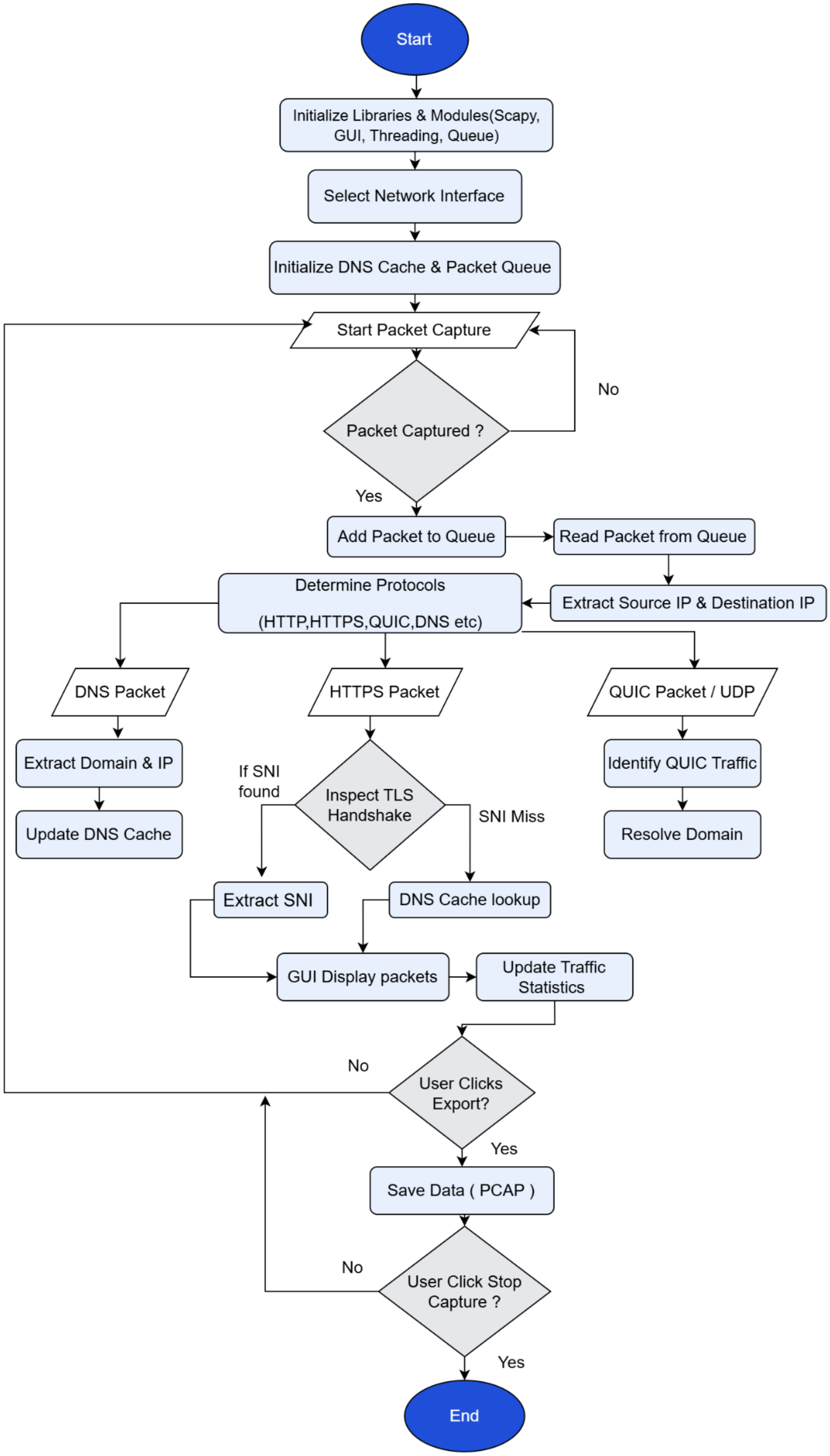


Figure 3.11: DNS-Based Network Traffic Analysis

3.4.1.2 URL Monitoring Using MITM Proxy

The second phase of monitoring is the application-layer HTTPS monitoring. This is achieved using a transparent MITM proxy, built using mitmproxy, to intercept encrypted HTTPS traffic, after TLS decryption, and parse full URLs and request information (metadata) in real-time. The following information are extracted from every intercepted request: Full URL, Domain name, HTTP method, Content type, Request path, Response information; the following Special processing are done for CDN platforms (like youtube) where domain names of CDNs are mapped to the origin website, to overcome tracking using domains of CDN:Duplicate, advertisements, analytic and background requests are dropped; all the urls and meta information collected are further given to AI filtering engine for ML classification and semantic analysis.

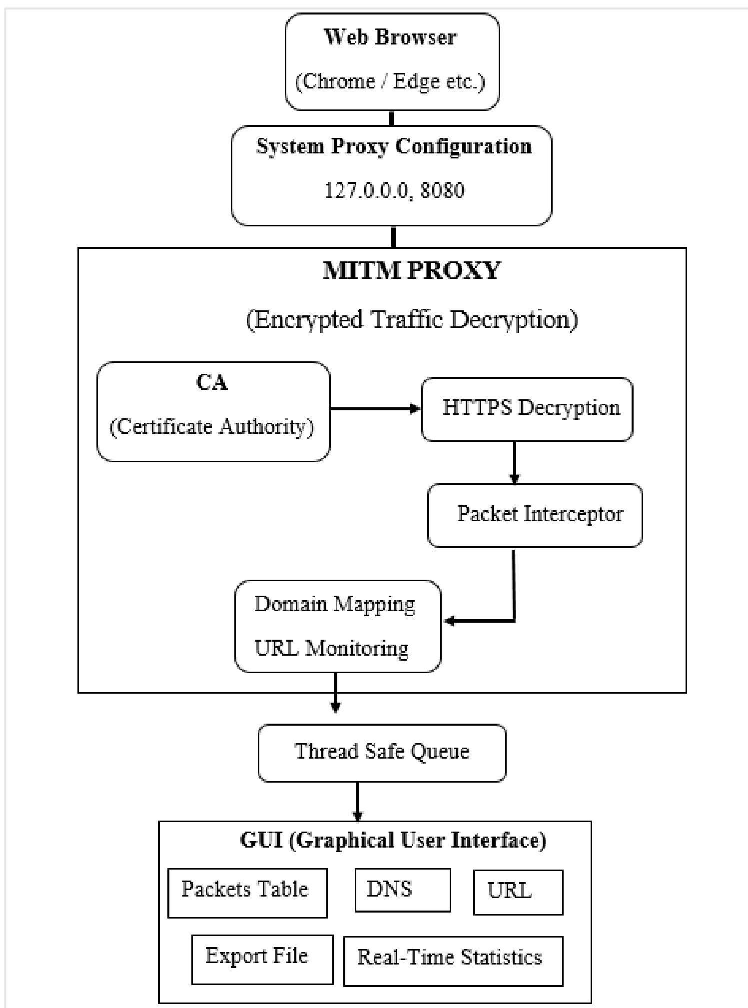


Figure 10: URL Monitoring Using MITM Proxy

Algorithm:

URL Monitoring Using MITM (Proxy)

- First of all, you fire up the application and load the necessary libraries to handle HTTPS interception, process packets, manage threads, and create the GUI.
- Next, you set up the GUI and prep your internal data structures think packet queues, storage buffers, stats counters, and export handlers.
- Then, check if the MITM proxy security certificate is installed. If it's not there, alert the user and show them how to get the cert set up along with configuring the browser proxy settings.
- The app waits for the user to start the packet capture through the GUI.
- Once capture starts, you spin up the HTTPS proxy server on port 8080 using MITM proxy tech.
- Now, you need to point the browser to direct HTTPS traffic through this local proxy to catch the encrypted stuff.
- From here, you keep intercepting HTTPS requests, but you don't mess with the data.
- For each capture, you grab metadata like the request method, full URL, host domain, path, scheme, headers, and timestamp.
- You filter out background, analytics, ad, and tracking requests to make things less noisy.
- To figure out the main page URL for sub-resource requests, you use the Referer header and request patterns.
- For CDNs, you match them back to their primary site domains.
- You work out the content type from HTTP headers, URLs, and methods too.
- To avoid dups, you compare new requests against recent ones and skip the repeats.
- Also, grab the response info like status codes and sizes whenever possible.
- All this info goes into a structured packet record.
- Send that record to a shared queue for the GUI to display in real-time in a table.
- Visually highlight based on content type to make it easier to scan like HTML, JSON, videos, images, and error types.
- Keep track of total packets and top domains for quick stats.
- Individual packet details are accessible when users select entries in the packet list.
- And you've got options to export the data in PCAP, JSON, or CSV formats.
- Users can also clear the data and reset stats if needed.
- Traffic gets monitored until capture is stopped.

- Shut down the proxy server, switch off capture, restore app settings, and remind the user to revert their browser proxy settings.
- Finally, wrap everything up nicely by closing out threads, ending processes, and freeing system resources on app exit.

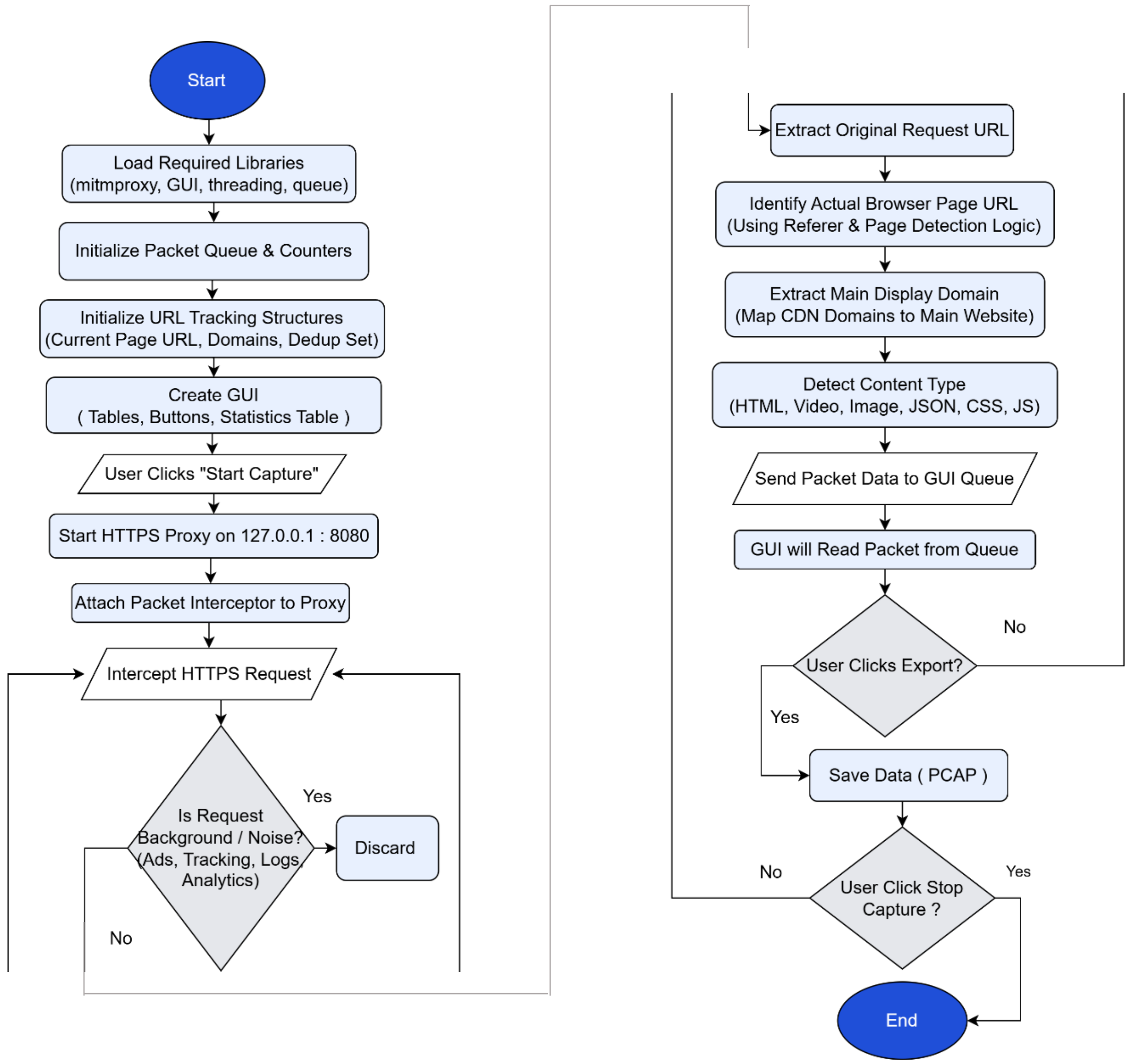


Figure 11: URL Monitoring Using MITM Proxy

3.4 Raspberry Pi Router Configuration

All the devices are monitored and filtered by this central system which is running on the Raspberry Pi 5 with the Raspberry Pi being set up as Wi-Fi router to receive internet via Ethernet and distribute access via Wi-Fi hot spot (hostapd & dnsmasq services).

We use IP forwarding and NAT configuration rules to achieve traffic redirection from and to network interfaces, to send them out to the Internet. We redirect HTTP and HTTPS traffic via an intermediary MITM transparent proxy using iptables rules to log and filter all internet traffic in real time.

With these router configurations, we can intercept, monitor and filter the internet traffic for all the users without having to install anything.

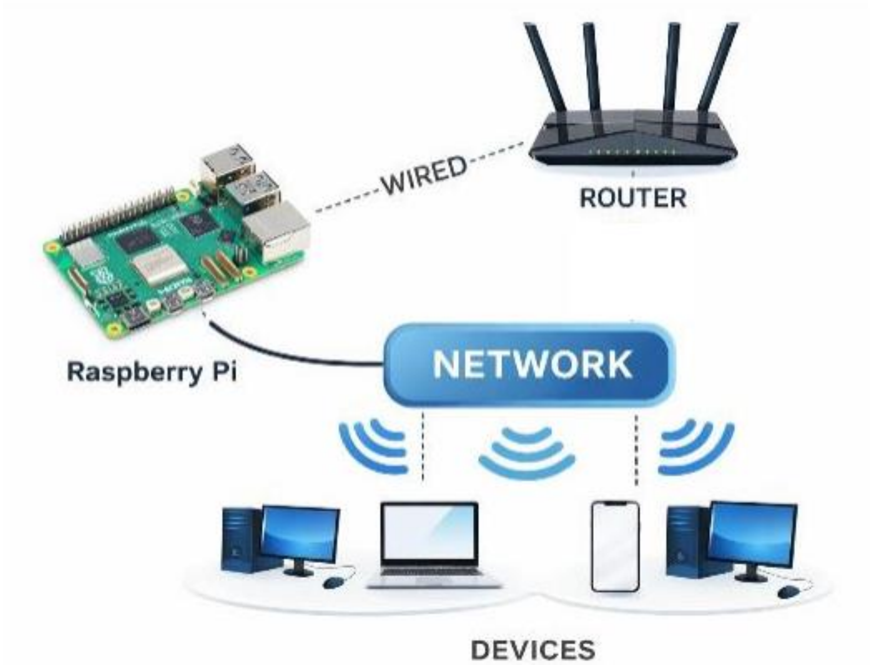


Figure 12: Raspberry pi5 Router Configuration

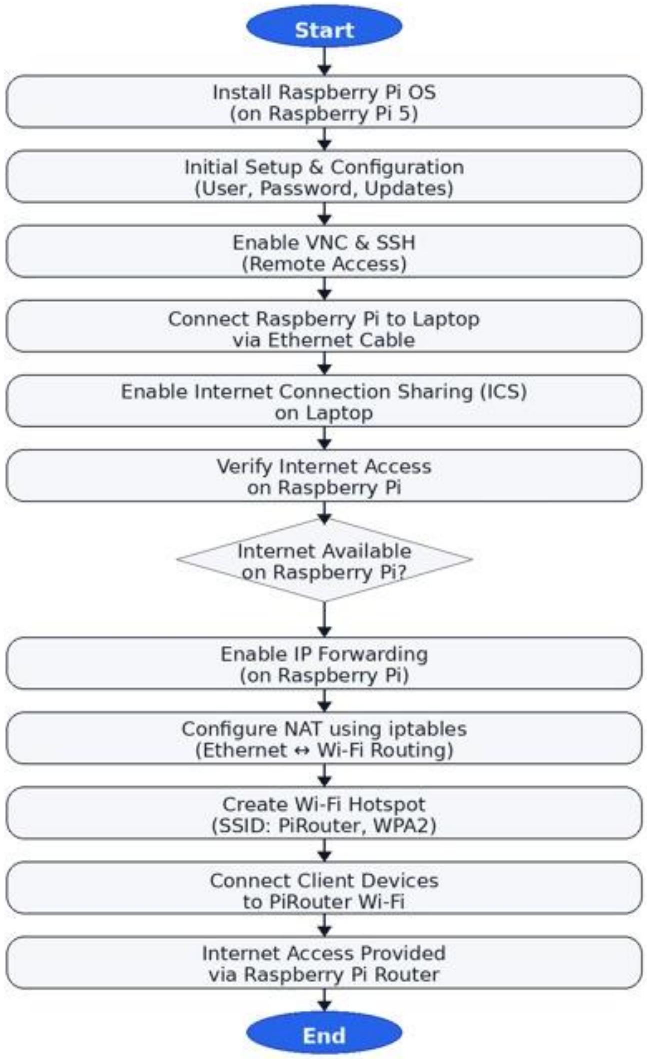


Figure 13: *Raspberry pi5 Router Configuration*

3.6 AI-Based Filtering and System Deployment

Once the monitoring modules and router is configured, the recorded traffic data then passed through the AI filter which aimed at real-time blocking of harmful and addictive content on the internet.Different activities online such as YouTube streaming, YouTube Shorts, Instagram Reels, TikTok and Gaming activities,educational content, and inappropriate searches were captured to train a dataset to enable intelligent classification.

We performed feature extraction of the collected traffic data like URL patterns, domain properties, request paths, query parameters, and traffic behavior. Many ML models like Logistic Regression, SVM, Random Forest, Gradient Boosting, XGBoost and CatBoost were trained and tested on the Google Colab and the best one (CatBoost) among the tested models was used to deploy the system.

In addition, a RoBERTa-based content classification with PyTorch has been added to more accurately grasp context and screen semantic contents of content.If the system is accessed in relation to the multimedia content, such as YouTube videos, the system gets all the metadata that refers to video title, description, channel and tags from YouTube Data API, and checks what kind of the content it refers to, and classifies the inappropriate, addiction or harmful content more accurately.

Furthermore, an intelligent signal scoring mechanism is provided for classifying inappropriate and safe content signals. The harmful signals are Adult Content, Gaming Addiction, Love Story content, Movie, Drama, etc. These signals are compared with safe educational and Islamic content signals in order to provide an intelligent filtering decision.

The system we designed implements a multi-layer intelligent filtering architecture composed of domain blacklisting, URL keyword analysis, smart signal scoring and RoBERTa based semantic analysis. Such architecture helps to boost filtering performance and achieve intelligent understanding of online context.

The whole filtering system is run on Raspberry Pi 5. Since the whole filtering system is hosted on the Raspberry Pi 5, it is able to monitor, analyze, and filter the Internet traffic of all connected devices at the same time.

3.7 AI-Based Blocking and Smart Redirection

The real time AI-based blocking mechanism works on the basis of the prediction from the filtering engine. Once inappropriate or addictive content is identified, the corresponding request is blocked automatically at the network level even before loading to user device.

Rather than a traditional 'blocked' or 'error' page, a smart redirect is employed to route the user toward beneficial content. For YouTube, the program blocks "Shorts", removes negative recommendations, blocks unsafe "thumbnails" and adds "positive" content like that educational and religious.

3.8 Mobile Application and Firebase Integration

To make the parents' monitoring and controlling screen portable and easily manageable, the KidSafeNet mobile application is built in Flutter. It makes use of Firebase services like authentication, real time monitoring, logging, cloud storage etc for its core features.

You can turn kid mode on and off from the mobile application, check what is being blocked, see the activity logs, change the filtering configurations, view real-time activity of the connected devices, etc. Through the mobile application. Using the kid mode, parents are provided with an immediate way to manage filtering configurations remotely through the mobile application. Once Kid Mode is enabled, all AI driven monitor, filter and blocking will be activated, once Kid mode is disabled all internet will be accessed directly.

Firebase Firestore service is employed to save blocked URLs, filtering statistics, configurations and logs that allow a real-time synchronization between the Raspberry Pi router and the mobile application. The Raspberry Pi constantly synchronizes with Firebase service to fetch new filtering configurations and send activity logs in real-time.

The combined features greatly enhance usability, remote management, and parental monitoring control of the system.

3.9 Summary

In this chapter, the approach and architecture of the KidSafeNet solution have been presented. The hardware as well as software components used for the design and implementation of the proposed solution were elaborated upon in this chapter. In addition to that, the concepts of traffic monitoring based on DNS, interception of HTTPS URLs via MITM proxy, router design based on Raspberry Pi, artificial intelligence-based filtering engine, and semantic analysis based on RoBERTa model have been introduced. Moreover, the use of Firebase and the Flutter application have also been discussed in this chapter.

Chapter 4

IMPLEMENTATION AND RESULTS

This chapter illustrates the implementation and testing of the AI Based Parental Control Router. It is developed in multi steps and integrated on Raspberry Pi 5 device which provides network traffic monitoring, data-set generation, ML model training and real-time content blocking and re-direction.

4.1 System Overview

The KidSafeNet system proposed was built and successfully deployed on Raspberry Pi 5 as an AI based parental control router integrating DNS monitoring, HTTPS URL interception, machine learning based filtering, semantic AI analysis, real-time blocking, smart redirection, and mobile based parental control.

Internet traffic of the linked devices passes through the Raspberry Pi router and gets checked, filtered and monitored prior to delivery to the destination. Inappropriate and addictive contents can be detected by the system, unsafe contents blocked, and users redirected towards safe learning materials. Firebase services and a Flutter based mobile application offers real-time monitoring, logs, and remote controls to parents.

4.2 Hardware Setup and Router Configuration

The hardware implementation consists of Raspberry Pi 5, a Raspberry Pi case with cooling fan, power supply, microSD card, and Ethernet connectivity. Raspberry Pi was configured as a Wi-Fi router to provide internet access to connected devices while simultaneously performing traffic monitoring and filtering operations.

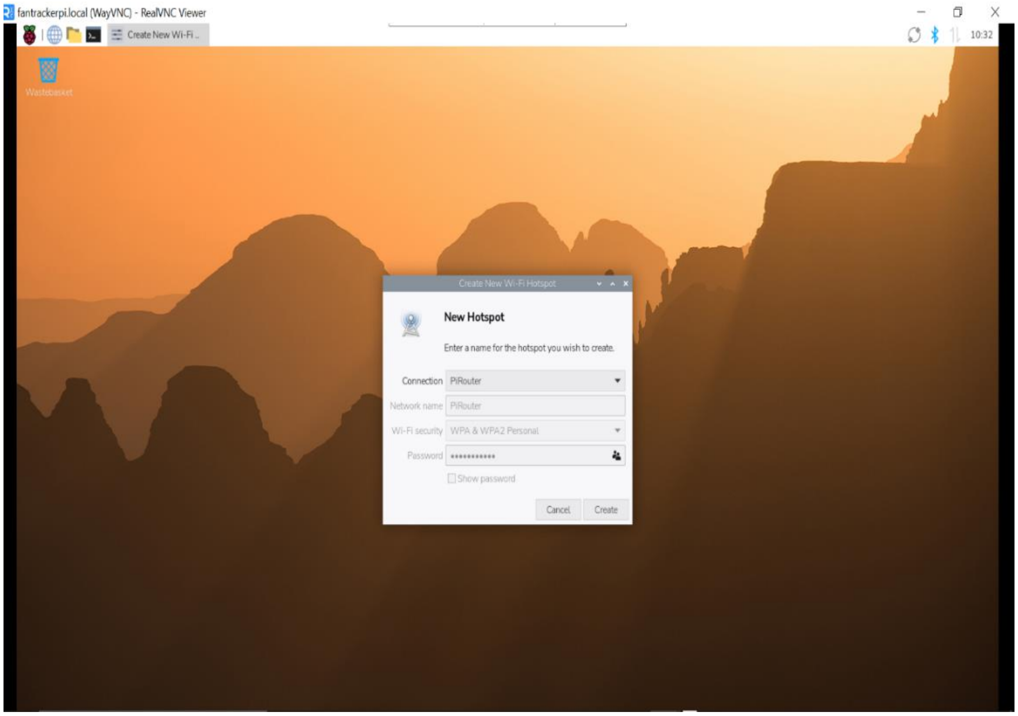


Figure 14: Raspberry Pi Configured as Router

Figure 15: Router Configuration and Device Connectivity

The following images show the full hardware set-up and working Raspberry Pi router. The devices were able to be hooked up to the router and then monitored centrally and protected with no software required to be put on each individual machine.

The setup was tested with various clients (e.g. Phone, laptop). Internet access was given to all connected devices through the Raspberry Pi hotspot. Because all network traffic is routed through the Raspberry Pi it is able to monitor and block the traffic from all devices centralized. This way it is easier for the parental control, and the user is not able to use the network from other devices.

4.3 DNS and URL Monitoring Implementation

The DNS & HTTPS monitoring system was built in Python by using Scapy and mitmproxy. In these modules, we have recorded Domain Names, IP Addresses and the full URLs for encrypted web traffic dynamically.

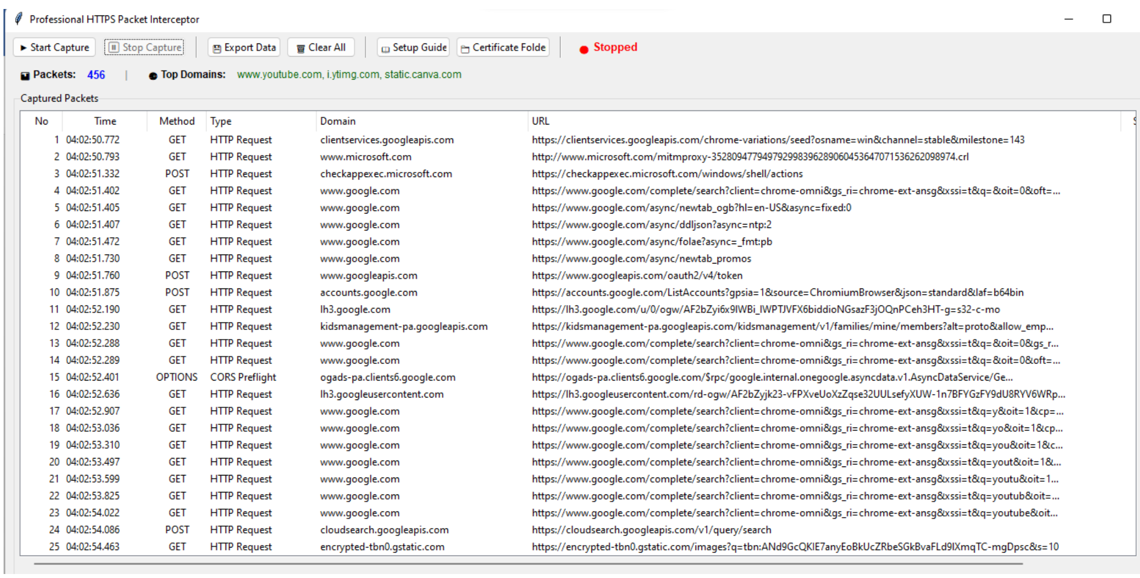


Figure 16: DNS and URL Monitoring Interface

The system effectively logged the browsing activities and extracted valuable traffic logs to feed the content analysis/filtering to the AI system. Extracted domain and URLs were again processed in the filtering engine for decision.

The module of monitoring of the DNS, recorded the sites and their IPs in real-time. The monitoring of the HTTPS URL provided more details, since it showed all the URL of encrypted traffic by means of the proxy MITM. As was possible to test, during the tests, it registered perfectly the user navigation in all the sites tested (school sites, search engine sites, social media, video streaming, etc.), and this information became the basis of our classification based on Artificial Intelligence and filtering.

4.4 AI-Based Filtering and Content Control

After collecting the traffic, artificial intelligence methods were applied to the data collected for the purpose of classifying and filtering the contents. A dataset was created based on the actual web surfing patterns. Several machine learning algorithms were trained using the generated dataset. Out of many classifiers, CatBoost was selected.

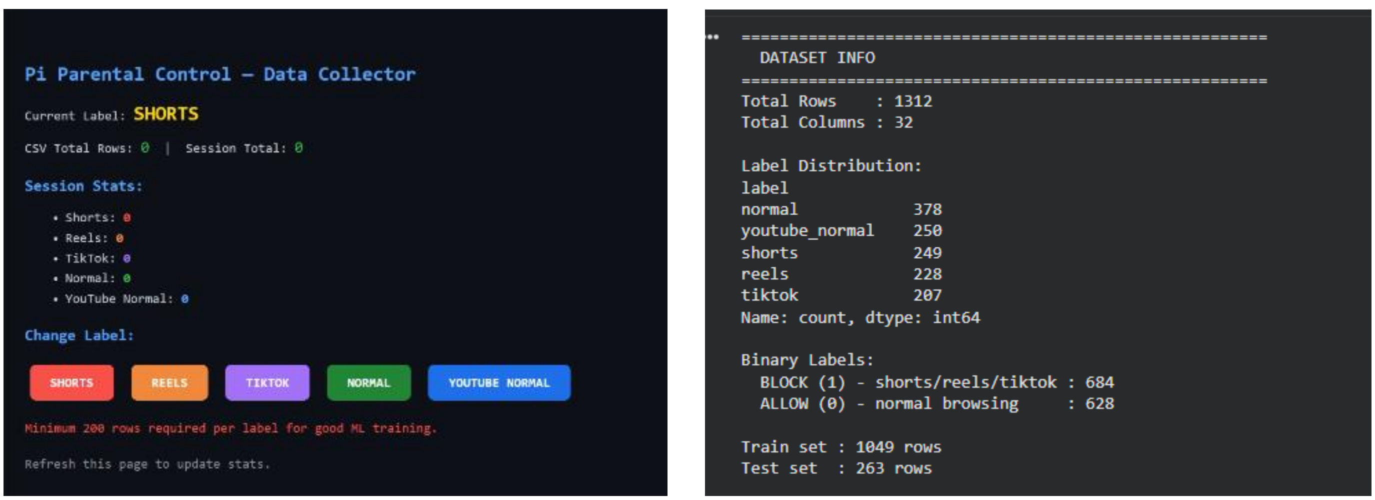
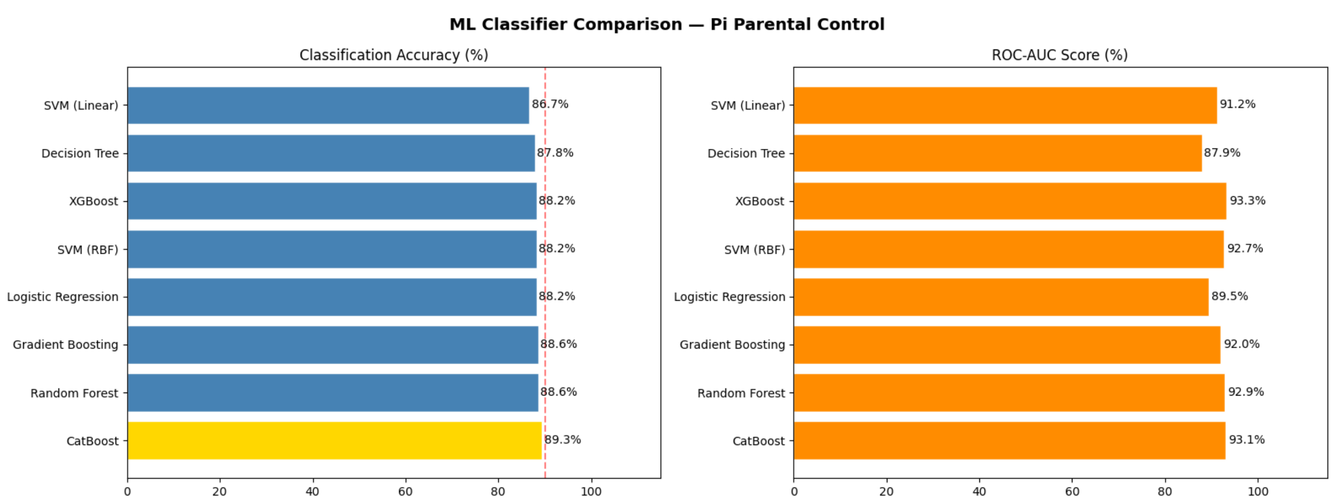


Figure 17: Dataset Generation

Several machine learning algorithms were tested against the data generated. Experiments conducted illustrated CatBoost performing at its highest level classification accuracy 89.35%, precision recall, and AUC scores. Confusion matrix reveals most safe and harmful browsing activities being classified correctly with limited false positive and false negative entries. Therefore, CatBoost is proven to be adequate to deploy for the developed parental control system in real time.

```
•••  ====================================
      TRAINING CLASSIFIERS
     ====================================

      Training: Logistic Regression...
        Accuracy   : 88.21%
        CV Score   : 87.80% ± 1.48%
        AUC        : 89.48%
        F1 Score   : 88.22%
        Precision  : 88.24%
        Recall     : 88.21%

      Training: SVM (Linear)...
        Accuracy   : 86.69%
        CV Score   : 88.66% ± 1.59%
        AUC        : 91.19%
        F1 Score   : 86.70%
        Precision  : 86.83%
        Recall     : 86.69%

•••   Training: Random Forest...
        Accuracy   : 88.59%
        CV Score   : 90.66% ± 1.36%
        AUC        : 92.89%
        F1 Score   : 88.59%
        Precision  : 88.59%
        Recall     : 88.59%

      Training: Gradient Boosting...
        Accuracy   : 88.59%
        CV Score   : 90.66% ± 1.18%
        AUC        : 92.01%
        F1 Score   : 88.60%
        Precision  : 88.61%
        Recall     : 88.59%

•••   Training: SVM (RBF)...
        Accuracy   : 88.21%
        CV Score   : 90.37% ± 1.54%
        AUC        : 92.73%
        F1 Score   : 88.22%
        Precision  : 88.29%
        Recall     : 88.21%

      Training: Decision Tree...
        Accuracy   : 87.83%
        CV Score   : 89.14% ± 2.96%
        AUC        : 87.93%
        F1 Score   : 87.83%
        Precision  : 88.09%
        Recall     : 87.83%

•••   Training: XGBoost...
        Accuracy   : 88.21%
        CV Score   : 90.75% ± 1.06%
        AUC        : 93.26%
        F1 Score   : 88.22%
        Precision  : 88.24%
        Recall     : 88.21%

      Training: CatBoost...
        Accuracy   : 89.35%
        CV Score   : 91.61% ± 0.87%
        AUC        : 93.06%
        F1 Score   : 89.36%
        Precision  : 89.37%
        Recall     : 89.35%
```

Figure 18: Model Performance Metrics (Accuracy & AUC)



```
=================================================================
Classifier                 Accuracy      CV Score       AUC
-----------------------------------------------------------------
🥇 CatBoost                   89.35%     91.61%±0.87    93.06%
🥈 Random Forest              88.59%     90.66%±1.36    92.89%
🥉 Gradient Boosting          88.59%     90.66%±1.18    92.01%
   Logistic Regression       88.21%     87.80%±1.48    89.48%
   SVM (RBF)                 88.21%     90.37%±1.54    92.73%
   XGBoost                   88.21%     90.75%±1.06    93.26%
   Decision Tree             87.83%     89.14%±2.96    87.93%
   SVM (Linear)              86.69%     88.66%±1.59    91.19%
=================================================================

🏆 BEST CLASSIFIER : CatBoost
   Accuracy        : 89.35%
   AUC Score       : 93.06%
```

Figure 19: AI-Based Classification Results

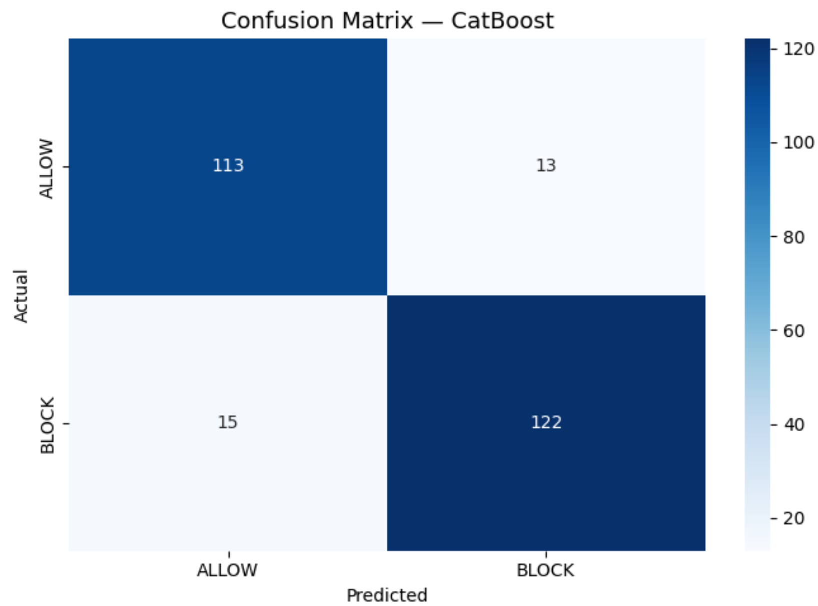


Figure 20: Confusion Matrix of CatBoost Model

For enhanced contextual awareness, content classification model with RoBERTa was added.Content classification model with the context will be able to classify the inappropriate or addictive content or any unsuitable material more efficiently. Filtering engine will make use of Machine Learning classification along with Semantic analysis and keyword based filter as well as smartly signals score for the smart content controlling.

Using RoBERTa-based content classification, system performance was further enhanced, which allow the system to understand online content based on contexts. Traditional keyword-based filtering is relying on exact word match. Semantic model is considering the meaning and intention of content to make classification. Combination of CatBoost classification and RoBERTa-based content classification has improved classification accuracy and intelligent control over content.

4.5 Shorts Blocking and Smart Redirection

An elaborate filtering system was built in order to detect and block YouTube Shorts and other addicting short form content, whenever the system finds Restricted Content automatically diverts to safe and educational contents.

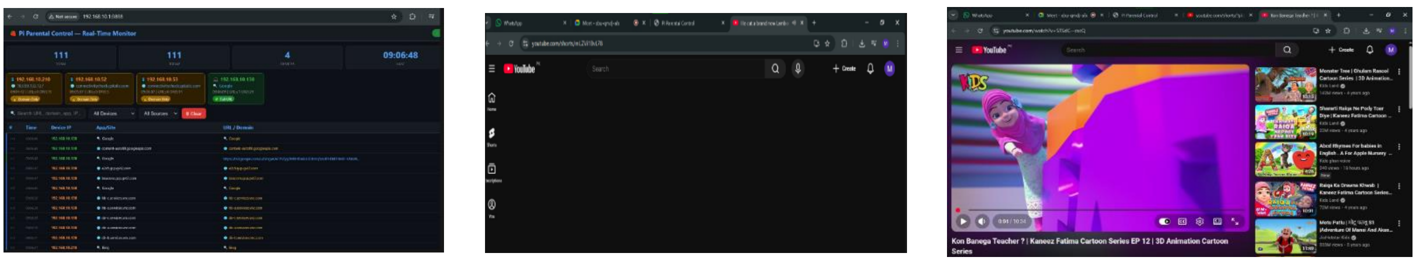


Figure 21: URL Monitoring, Shorts Blocking and Redirection

Results demonstrate YouTube Shorts to be successfully blocked while a browser is simultaneously redirected towards informative and kid-safe content to keep a user from viewing a time-killing source, yet a pleasant browsing session.

The results also show that YouTube Shorts URL were recognized and filtered at a level preceding the presentation of content. Where a harmful or addiction producing content was detected, the system will present to the user safe and educational web content automatically. It serves as a protection mechanism while keeping the child friendly web experience of the user.

4.6 Advanced Content Filtering and Platform Blocking

The system makes full use of semantic AI to filter contents through deep understanding and intelligent classification. Websites and applications like Tiktok, Instagram can be blocked on the network level. Parents set the filtering rules for children in the settings.

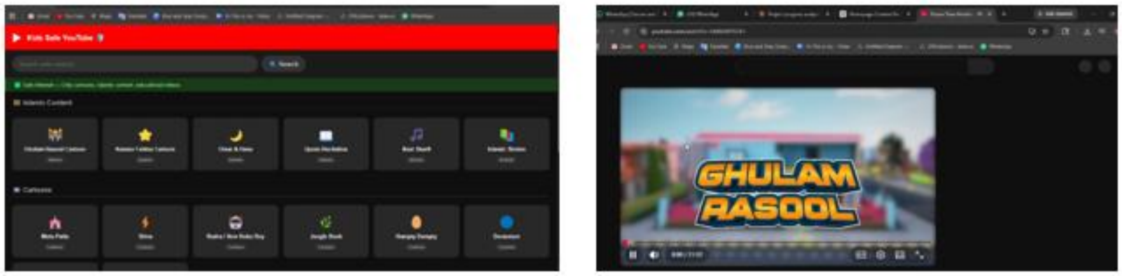


Figure 22: Advanced Content Filtering and Platform Blocking

It has achieved the goal of denying access to sites and programs with addictive and harmful contents and giving access to safe educational ones. The multi-layer filtering approach described has proven to be efficient.

Combining blacklist filtering, URL filtering, ML-based classification, semantic AI analysis, and signal-based scoring results in a layered filtering mechanism which is more precise and robust than typical rule-based parental control systems. The experimental results prove the layered mechanism effective in detecting harmful online content.

4.7 Mobile Application and Firebase Integration

A real-time parental control and monitoring mobile app was built in Flutter. The system used Firebase services for authentication, cloud sync, activity logging and system remotely managed from the parents end.

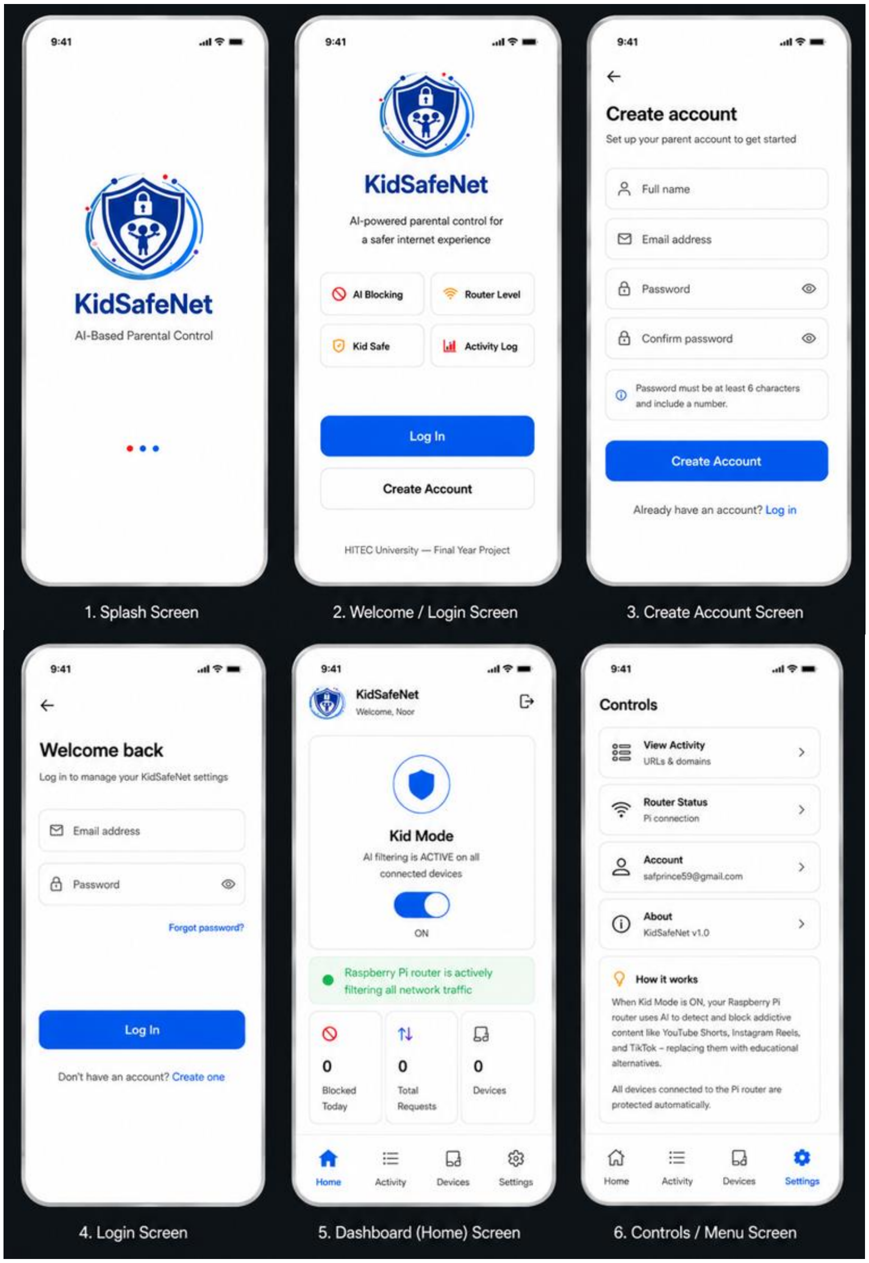


Figure 23: KidSafeNet Mobile Application

This feature lets the parents turn the kid mode ON/OFF, watch browsing history, check blocked pages and enable or disable content filters from anywhere in the world. Since, Firebase and Raspberry Pi were synchronized in real time, the policy of parental control can be immediately implemented on the network.

The test showed that Firebase syncing was able to update filtering configuration between mobile application and Raspberry Pi in real time. Parents can remotely enable or disable Kid Mode, check browsing activity history, blocked history and control filtering option in one place.

4.8 Overall System Evaluation

The outcomes show the successfully combine network monitoring, intelligent filtering, router deployment, cloud synchronized and mobile based parents control to form KidSafeNet system, detecting harmful and addicting network content, and blocking and guiding towards healthy and educating network environment.

The successful test on the Raspberry Pi 5 provides strong proof of concept for the feasibility, scalability and effectiveness of the above solution for current Parental Control applications and secure internet browsing.

4.9 Summary

The current chapter has outlined the implementation and performance testing of the proposed KidSafeNet framework. It has discussed how the Raspberry Pi routing system, the DNS and URL inspection modules, traffic classification using machine learning, and the semantic filtering algorithms have been implemented successfully. Based on experimental findings, it has been proven that the proposed CatBoost classifier and RoBERTa-based semantic filtering model are capable of efficiently detecting harmful and addictive content online. It has been highlighted that short form content filtering and smart redirection has been successfully implemented as well.

Chapter 5

ENVIORNMENT AND SUSTAINABILITY

As networking is encompassing more of a digital infrastructure, then a solution to meet the demand is the provision of the responsible future for a networking and digital infrastructure with networking is responsible solutions. The AI-based parental control router based on Raspberry Pi and based on open source is the idea of the green computing, responsibility with the digital infrastructure. This chapter explains how the system, implemented as the project's solution to internet safety, reduce the ecological footprint, in home, school, or even small office environments, with a sustainable and digital safe framework.

Through use of small and lightweight hardware and the efficient real-time network analysis with open-source tools for network monitoring it can reduce electronic wastage and power consumption, because it relies on efficient monitoring rather than an entire network switch, like the commercial counterpart with an embedded computer that handles large volumes of traffic at an advanced level, which has a larger power usage and footprint. It directly correlates with the relevant UN SDGs such as digital safety, justice and efficient infrastructure for sustainable and smart futures.

5.1 Alignment with Sustainable Development Goals (SDGs)

The presented system promotes some United Nations SDGs by increasing safety in the digital world, as well as a reasonable technology use and a sustainable computing environment. Specifically, the SDGs related to the project are:

- SDG 16: Peace, Justice, and Strong Institutions

All these aims are reinforced by the projects emphasis on transparency at the network-level,child protection,low power hardware, and open-source development.

5.1.1 SDG 16: Peace, Justice, and Strong Institutions

This project directly helps the SDG 16 and calls for an increase in the web security and justice. Children are one of the most fragile web users and they are prone to negative uses of excessive and addicting websites. This application developed system enables parent/guardians or educator to monitor their children's internet usage on a network level in a secure and responsible manner. Being able to check the domains and URLS in real time, it establishes transparency of online activity. It is not conducive of online safety for blind-folded blocking or redirecting. The transparent blocking system establish accountability and trust between parents/guardians/educators and children in the promotion of peace, safety, and justice.

5.2 Summary

This chapter explored the environmental and sustainability issues related to the KidSafeNet solution proposal. The role played by the project in promoting responsible use of the internet, digital safety, and sustainable computing has been emphasized. The relevance of the system to the UN Sustainable Development Goals, especially goal 16 (Peace, Justice, and Strong Institutions), was made clear. It was shown that the solution not only promotes digital safety for kids but is environmentally friendly in terms of hardware used and software technology.

Chapter 6

CONCLUSION

6.1 Lessons Learned:

The KidSafeNet-an AI-based parental control router was designed in this project to perform intelligent and real time internet filtering for kids. In this work the we managed to combine DNS monitoring, HTTPS URL interception, Machine learning, semantic AI based analysis and router level deployment in an all-in-one Parental control platform. The KidSafeNet application can identify the unsafe and addictive content on the internet, block the inappropriate content and redirect the user towards the safe educational sites. CatBoost-based machine learning and RoBERTa based content classification enhance the efficiency of content filtering and provides better understanding on the context of internet usage. It supports specific content filtering over YouTube shorts, Instagram reels, TikTok, etc. Furthermore, firebase integration along with flutter mobile application support in the KidSafeNet provides real-time monitoring on kids' activities, logs the kid's internet usage and supports remote control of the Kid Mode. The all-in-one system was deployed successfully over Raspberry Pi 5. It provides centralized control of the internet filtering for all the devices without installation over end-user device.

6.2 Future Work:

- Future improvements could involve AI for analyzing images, videos, and audio to boost filtering accuracy.
- Create a separate mobile app that works without a router connection.
- Age-based personalization and kid-specific profiles are on the list too.

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